



Proposed CEA Link — December 2026

How this rota would work from December 2026, and how it compares with today's link

Prepared 28 September 2026 · every figure is calculated from the rota, not entered by hand

PROPOSAL

Just Enough

Fifteen shift times, changed in the fewest places that meet all eleven rules

CODE **JE-24-M29**
FINGERPRINT **49717d70**
FAMILY Fifteen Turns
CREATED 28 September 2026

- ✓ **24** lines, one per person
- ✓ **35h 00m** a week Mon–Sat, on average — exactly the contract
- ✓ longest run **9** days (limit 13)
- ✓ **0** rests under 12h
- ⚠ **3** design-specific fatigue findings (today 5)
- ✓ all **11** current working rules met
- ⚠ Sunday's finish undecided — 5 trains after 23:25

Five questions a manager asks first, in order. Every figure is worked out from the rota; **page 2 says how to read each one**, with today's link and the other proposals marked on a scale.

CAN IT BE RUN?

Yes

meets every hard limit

0 rests under 12h · longest run 9 days (limit 13) · contract exact

DOES IT MEET THE DECEMBER RULES?

11 of 11

rules met

current working rules, given verbally and not yet in writing (page 7) · today's link meets 4 of 11

HOW TIRING IS IT?

3

design-specific fatigue findings

plus FF2 and FF18, standing on every weekly link and not counted (page 8) · today 5 · fewest of the others 0

DOES IT FOLLOW THE TRAINS?

34.3

how closely the floor follows the weekday trains — **0 is a perfect match**

today 52.0 · Sat 51.4 (today 65.9) · Sun 34.9 (today 71.7) · best of the others 26.5

IS IT FAMILIAR?

5 of 15

shift times already worked today

10 new, listed on page 5 · 9 of 20 working weeks keep one shift time (today 7 of 16)

2 in 24

full weekends off

8% of the rotation, against 20% today (4 in 20) · most of the others 25%

4 cover weeks

lines 6, 12, 18, 24

evenly spaced, every 6 lines · today's are at lines 1, 7, 12, 17

2h 26m a week

typical week-to-week move

how much the start time typically changes from one week to the next — today's link moves 4h 00m a week

What this is

Just Enough is a proposal for the CEA link on the December 2026 timetable: a 24-line rotation for 24 people, where today's link has 20. In short: fifteen shift times, changed in the fewest places that meet all eleven rules.

Against today's link — the table below; page 2 explains each figure.

AT A GLANCE — GREEN IS BETTER THAN TODAY, AMBER IS WORSE, UNSHADED IS NEITHER	TODAY'S LINK	JUST ENOUGH
December 2026 working rules met (given verbally — page 7)	4 of 11	11 of 11
People working each day — weekday · Saturday · Sunday	11–12 · 10 · 8	14–15 · 14 · 10
How closely the floor follows the trains — weekday · Sat · Sun (lower is closer)	52.0 · 65.9 · 71.7	34.3 · 51.4 · 34.9
Design-specific fatigue findings (page 8; FF2 and FF18, on every weekly link, are not counted)	5	3
Most days worked in a row (limit 13)	9	9
Shortest rest between two duties (at least 12 hours)	12h 30m	14h 05m
Full weekends off	4 in 20	2 in 24
Different shift times	18	15 — 5 already worked today

What it is not. Not a recommendation or a finished roster: nothing here has been through the roster office or a union rep yet.

HOW TO READ THE NUMBERS

What each figure means, and what good looks like

Every figure on this sheet is worked out from the rota. On each scale: this sheet in navy, today's 20-line link as a ring, the best and worst of the other 27 proposals as gold and grey ticks.

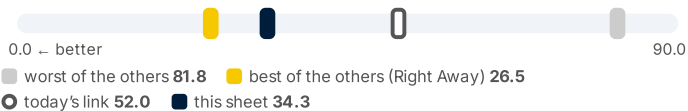
Three things to keep in mind. Lower is better on the fit, the fatigue count and the days in a row; higher is better on rest, weekends, weeks on one shift time and familiarity. "Present" is not "failed" — the ORR list reports what is there. Today's link is the yardstick, not the target: it meets 4 of the 11 December 2026 rules.

DOES IT FOLLOW THE TRAINS?

How well the floor follows the trains — weekday

For each hour of an average weekday, the share of the day's floor staff on duty is set against the share of the day's trains — an hour with a tenth of the trains should have about a tenth of the staff. 0 would be a perfect match; a typical hour here is about 1.4 percentage points off, against 1.7 today. The ticket office is left out, bar its second person's help at the quiet ends.

What it does not tell you. It compares shares, not numbers: an extra person in a quiet hour makes the score worse though nobody is worse off. Read it beside the hour-by-hour numbers on page 6.



Saturday **51.4** (today 65.9) · Sunday **34.9** (today 71.7). Everyone on duty, ticket office included: weekday 29.0 (today 50.2).

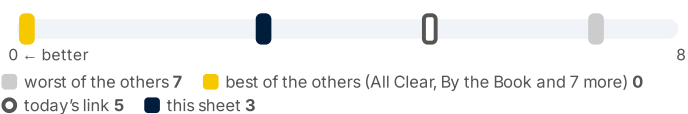
On page 6.

HOW TIRING IS IT?

Design-specific fatigue findings

How many of 25 roster patterns appear anywhere in the rotation — long runs of earlies, short gaps between duties, start times that jump. 21 are on the Office of Rail and Road's good-practice list; 4 are extra checks from the rail industry's fatigue guidance. Two come with every weekly link — FF2 (the 06:20 open) and FF18 (a weekly rotation) — so page 8 records them and they are not counted.

What it does not tell you. A factor present is worth a look, never a breach: the ORR says the list is guidance, not limits. Zero does not mean approved and one does not mean unsafe; page 8 names which, and how big.



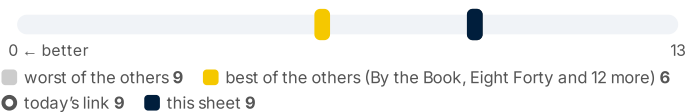
On page 8.

CAN IT BE RUN?

Most days worked in a row

The most days in a row anyone works in the worst case — a cover week's four duties placed as badly as they can be. Chiltern's limit is 13; the Links designer aims for six or fewer.

What it does not tell you. The roster clerk decides where a cover week's four duties fall; page 7 says whether that changes this figure on this design.



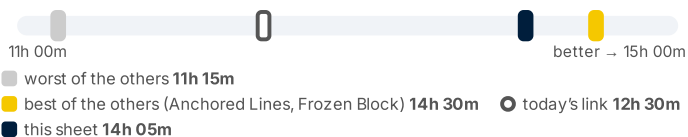
On page 7.

CAN IT BE RUN?

Shortest rest between two duties

The shortest gap anywhere between one duty ending and the next beginning — Saturday into Sunday and the last line into the first included. Twelve hours is the minimum; a design with less cannot be run.

What it does not tell you. It is one gap. A rota with many rests of just over 12 hours and one with none look the same here; the number under 12 hours is at the top of page 1.



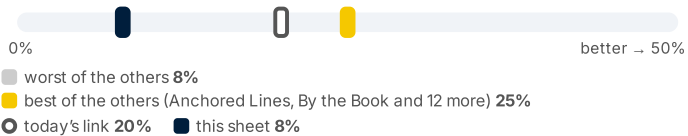
On page 7.

WHAT IS IT LIKE TO WORK?

Full weekends off

The share of weekends off: a Saturday off followed by a Sunday off at the start of the next line. Today's link gives 4 in 20, one weekend in 5; this sheet gives 2 in 24, one weekend in 12.

What it does not tell you. A weekend off beside a 42-hour week is still a weekend off; the hours by week are on the grid, page 4.



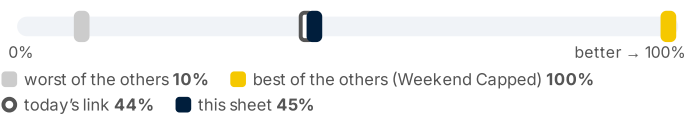
On page 9.

WHAT IS IT LIKE TO WORK?

Working weeks on one shift time

The share of working weeks that are one-turn weeks: the same clock time Monday to Friday, and every day of the week an early or every day a late. A week that mixes an early and a late is what people mean by "all over the place".

What it does not tell you. One shift time all week can still be a hard week if the shift is long; the week-to-week move and the duty lengths are on pages 5 and 8.



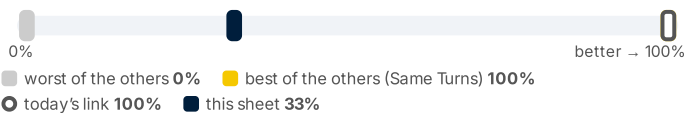
On page 5.

IS IT FAMILIAR?

Shift times people work today

The share of this sheet's clock times that somebody on the 20-line link already works. It is a rough guide to how much there is to learn and get used to.

What it does not tell you. Familiar is not popular, and a new time in one week is not one in ten. Page 5 lists the new ones.



On page 5.

The decision this sheet supports

What is being decided, which criteria are fixed and which are yours to weigh, how much each figure can bear, and what no page here can tell you. Read this before the figures.

- 1 What this is
- 2 How to read the numbers
- 3 The decision it supports (this page)
- 4 The 24-line grid, with the words explained
- 5 A week, and the duty table
- 6 People on duty, hour by hour
- 7 Hard limits and the December 2026 rules
- 8 The ORR fatigue factors
- 9 Beside the other proposals
- 10 The rota, ready to paste into the app

The question

Should the CEA link for the December 2026 timetable be **Just Enough**? It is judged against today's 20-line link and the December 2026 rules, and page 9 sets it beside the other 27 proposals on the same figures.

Just Enough — fifteen shift times, changed in the fewest places that meet all eleven rules.

The criteria, in three tiers

HARD — A DESIGN MEETS THESE OR CANNOT BE RUN

- ✓ Twelve hours between duties — met, tightest 14h 05m
- ✓ Chiltern's 13-day limit — longest run 9 days
- ✓ The 35-hour contract, exactly — 35h 00m

SOFT — WEIGH THESE AGAINST EACH OTHER

- December 2026 rules met: **11 of 11** (today 4 of 11)
- Design-specific fatigue findings: **3** (today 5)
- How the floor follows the trains, weekday · Sat · Sun: **34.3 · 51.4 · 34.9** (today 52.0 · 65.9 · 71.7; lower is closer)
- Times worked today: **5 of 15** · full weekends off: **2** in 24 (today 4 in 20)

PREFERENCE — FOR STAFF TO SAY, NOT THE APP

- How long the shifts are: earlies 7h 00m–8h 30m, lates 7h 30m–8h 30m
- 10 of 15 clock times are new — the duty table on page 5 lists them
- The longest duty, 8h 30m — the grid on page 4 shows which lines carry it
- Cover weeks at lines 6, 12, 18, 24

How much to trust each figure

FIGURE ON THESE PAGES	WHERE IT COMES FROM	WEIGHT
0 rests under 12h · 35h 00m contract · 14 on Saturday, 10 on Sunday	Worked out from the rota; checked again by the app when it is loaded	Firm
Longest run 9 days, inside 13	Computed; the written source of the 13-day limit is still to be confirmed	Firm
At least four at the open, three at the close, five at 22:00; 14 on Saturday, 10 on Sunday	December 2026 staffing levels given verbally; not yet in writing	Provisional
3 fatigue factors present	The ORR's 2021 list (21 of them) and four rail-industry checks — guidance, not limits; 1 factor whose exact definition is still to be confirmed	Advisory only — never pass or fail
Fit, everyone on duty 29.0 · 24.0 · 36.2; floor only 34.3 · 51.4 · 34.9	The measured December 2026 timetable; one way of measuring, not the only one; the floor figure leaves the office out (page 6)	A guide
5 of 15 times are today's	Compared with today's 20-line link	Firm , but only a rough guide to what staff will accept

What no page here can tell you

- Whether staff accept the 10 shift times nobody works today (page 5 lists them) — this sheet counts times already worked, which is only a rough guide to what people will accept.
- Where the roster clerk places a cover week's four duties. It changes the longest stretch without a two-day break (FF11, page 8): 10–11 if the four are worked together, 13 if they are spread out. The rota itself does not decide it.
- How leave and sickness cover land on a 24-line link with four cover weeks (lines 6, 12, 18, 24); nothing here models absence.
- Whether this pattern holds on Wembley event days — it is based on the train timetable, not on passenger numbers.

Before it is agreed

18 of the 20 working lines (lines 1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 14, 15, 16, 17, 19, 20, 21 and 22) hold at least one shift time nobody works today. On a rotating link everyone works every line in turn, so staff read those weeks first and say in one line each what they would change. The roster office answers the cover-week question. Someone confirms the source of the 13-day limit. Until then, this sheet is an aid to discussion, not a decision.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat	TURNS Mon-Fri
1	RD	06:20 13:20	06:20 13:20	06:20 13:20	06:20 13:20	15:45 23:55	RD	36h 10m	36h 10m	5	2
2	07:15 15:45	06:20 13:50	RD	RD	06:20 13:50	06:20 13:50	06:20 14:50	31h 00m	39h 30m	4	1
3	RD	RD	06:20 13:50	06:20 13:50	06:20 13:50	06:20 13:50	RD	30h 00m	30h 00m	4	1
4	07:15 15:45	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	RD	15:55 23:55	40h 00m	48h 30m	5	1
5	RD	RD	07:00 15:00	07:00 15:00	07:00 15:00	07:00 15:00	14:30 22:00	39h 30m	39h 30m	5	1
6	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
7	RD	RD	07:00 15:00	07:00 15:00	07:00 15:00	07:00 15:00	06:20 14:50	40h 30m	40h 30m	5	1
8	07:15 15:45	07:00 15:00	RD	RD	07:00 15:00	07:00 15:00	07:00 15:00	32h 00m	40h 30m	4	1
9	10:00 18:30	13:00 21:30	13:00 21:30	13:00 21:30	RD	RD	06:20 14:20	33h 30m	42h 00m	4	1
10	RD	13:00 21:30	13:00 21:30	13:00 21:30	13:00 21:30	13:00 21:30	RD	42h 30m	42h 30m	5	1
11	RD	06:20 14:20	06:20 14:20	15:45 23:55	15:45 23:55	14:00 22:30	RD	40h 50m	40h 50m	5	3
12	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
13	14:00 22:30	14:00 22:30	RD	RD	14:00 22:30	14:00 22:30	14:30 22:00	33h 00m	41h 30m	4	1
14	15:25 23:25	15:45 23:55	15:45 23:55	RD	RD	15:00 23:00	15:00 23:00	32h 20m	40h 20m	4	2
15	RD	RD	15:45 23:55	15:45 23:55	15:45 23:55	15:45 23:55	16:25 23:55	40h 10m	40h 10m	5	1
16	15:25 23:25	15:45 23:55	15:45 23:55	15:45 23:55	RD	RD	07:00 15:00	32h 30m	40h 30m	4	1
17	14:00 22:30	RD	RD	RD	06:20 14:20	06:20 14:20	15:55 23:55	24h 00m	32h 30m	3	1
18	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
19	RD	06:20 13:50	06:20 13:50	06:20 14:20	13:00 21:30	13:00 21:30	13:00 21:30	48h 30m	48h 30m	6	3
20	RD	RD	14:00 22:30	14:00 22:30	15:45 23:55	15:45 23:55	15:55 23:55	41h 20m	41h 20m	5	2
21	15:25 23:25	15:45 23:55	RD	RD	RD	06:20 14:20	06:20 14:20	24h 10m	32h 10m	3	2
22	07:15 15:45	07:00 15:00	07:00 15:00	07:00 15:00	RD	RD	RD	24h 00m	32h 30m	3	1
23	RD	14:00 22:30	14:00 22:30	14:00 22:30	14:00 22:30	RD	RD	34h 00m	34h 00m	4	1
24	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—	—
On duty	10 SP 4	14 SP 4	15 SP 4	14 SP 4	15 SP 4	15 SP 4	14 SP 4	35h 00m	38h 29m	4.35	9 one-turn weeks
Minutes	5,010	6,750	7,230	6,780	7,230	7,290	6,720	42,000	47,010		

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines · Turns: how many different shift times the week works Monday to Friday; **bold** marks a one-turn week (see below) · Minutes: total duty minutes each day — Monday to Saturday adds up to the contract (20 people × 35 hours = 42,000 minutes)

The words on this page

Link — the whole rota: 24 weeks laid out as 24 lines. Everyone works line 1, then line 2, and so on round the wheel; the link needs 24 people.

Turn — a shift time, such as 06:20-13:20. An **early** starts before 11:00; a **late** starts at 11:00 or later.

Cover (spare) week — a week with no fixed times. You work four duties of the seven days, placed by the roster clerk to cover leave and sickness.

The contract — 35 hours a week, averaged over the 24 lines; a cover week counts as a full contracted week.

The floor — staff out on the station (gates, concourse, platforms); the ticket office joins it only at the quiet ends (page 6).

Handover — the overlap when one shift takes over from another.

Line — one week of the pattern, Sunday to Saturday. You move down a line each week, and line 24 goes back to line 1.

Rest day (RD) — a day off inside the contracted week. Two together make a proper break; one on its own does not count as a 48-hour break.

Sunday — not part of the contract. A Sunday duty is rest-day working, paid on top of the week, which is why the hours are shown Mon-Sat and all week.

One-turn week — one clock time Monday to Friday, with every day of the week an early or every day a late (shown bold in the Turns column); the easiest kind of week to live around.

Opener / closer — the first people on duty in the morning and the last at night.

Fit — how closely the number of people on duty follows the number of trains through the day; 0 is a perfect match, lower is better.

A week, and the duty table

How a working week compares with today's, and every shift time beside the ones worked today.

The shape of a week, against today

MEASURE	TODAY (20)	PROPOSED (24)
Weeks on one shift time	7 of 16	9 of 20
Weeks mixing early and late shifts	7	9
Rest-day breaks of two days or more (of all rest-day breaks)	13 of 17	16 of 23
Single rest days between duties (not a 48h break)	4	7 (3 beside a cover week)
Days worked in a week (Sunday included; weeks × days)	1×3, 3×4, 11×5, 1×6	5×4, 13×5, 2×6
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 6, 12, 18, 24
Different shift times (10 not worked today)	18	15
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted)	8 of 16 lines	10 of 20 lines

How many people are working

PEOPLE WORKING	TODAY	PROPOSED	WHY
Monday to Friday	11–12	14–15	24 people instead of 20, so more on each weekday
Saturday	10	14	meets the December 2026 figure of 14
Sunday	8	10	meets the December 2026 figure of 10
Ticket office Mon–Fri · Sat · Sun	4 · 4 · 3	4 · 4 · 4	ticket office only; today Sunday has one late, not two
Opening at 06:20 (07:15 Sunday) Mon–Fri · Sat · Sun	4 · 4 · 3	4–5 · 4 · 4	at least four to open, every day — met
Through to the close (23:55; 23:25 Sunday) Mon–Fri · Sat · Sun	2 · 2 · 3	3 · 4 · 3	at least three to close, every day — met
Still on duty at 22:00 Mon–Fri · Sat · Sun	4 · 2 · 3	5–6 · 5 · 5	at least five from 22:00, every day — met

The duty table, beside today's

SHIFT TIME	TODAY'S LINK			PROPOSED		
	MON–FRI	SAT	SUN	MON–FRI	SAT	SUN
06:20–13:20 7h 00m · Early				1 (+1)		
06:20–13:35 7h 15m · Early	±			0 (-1)		
06:20–13:45 7h 25m · Early	±			0 (-1)		
06:20–13:50 7h 30m · Early				2 (+2)		
06:20–14:00 7h 40m · Early		±			0 (-1)	
06:20–14:20 8h 00m · Early	2			2 (+2)		
06:20–14:50 8h 30m · Early		3		2 (-1)		
07:00–15:00 8h 00m · Early				3 (+3)	2 (+2)	
07:15–15:45 8h 30m · Early			3			4 (+1)
08:00–16:30 8h 30m · Early	±			0 (-2)		
08:30–16:30 8h 00m · Early			±			0 (-1)
10:00–18:30 8h 30m · Early						1 (+1)
11:00–19:30 8h 30m · Late	±			0 (-1)		
12:00–20:00 8h 00m · Late		±			0 (-1)	
12:00–21:00 8h 00m · Late			±			0 (-1)
13:00–21:30 8h 30m · Late				2 (+2)	1 (+1)	
12:30–21:00 7h 30m · Late		±			0 (-1)	
12:30–22:00 8h 30m · Late	±			0 (-1)		
14:00–22:30 8h 30m · Late	2					2 (+2)
14:30–22:00 7h 30m · Late		2		2		
14:30–23:25 8h 55m · Late			±			0 (-3)
14:45–23:55 9h 10m · Late		±		0 (-2)		
15:00–23:00 8h 00m · Late				1 (+1)	1 (+1)	
15:15–23:55 8h 40m · Late	±			0 (-2)		
15:25–23:25 8h 00m · Late						3 (+3)
15:45–23:55 8h 10m · Late				3 (+3)		
15:55–23:55 8h 00m · Late				3 (+3)		
16:25–23:55 7h 30m · Late				1 (+1)		

People on each shift time on a Saturday, a Sunday and — under Mon–Fri — the weekday with most of that time, so Mon–Fri can add up to more than work on one day. Green: more than today; amber: fewer; struck through: not used.

Monday to Friday add up to 35,280 minutes of duty and Saturday 6,720 — together exactly 20 people × 35 hours = 42,000 minutes, the contract. Sunday adds 5,010, paid on top of the contracted week. 10 of the 15 shift times are new — nobody works them today.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's trains.

How to read it: each blue square is the number of people on duty in that hour (darker blue = more people). The orange row is the timetable — the train carriages arriving and leaving in that hour, which is how busy it is (darker orange = busier). Read the columns top to bottom: cover should be thickest where the orange is darkest. The **fit** figure at the right end of each row scores that match for the whole day.

Monday to Friday

staffed 06:20–23:55 · a row for each different weekday

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY MON · WED		4	4	6	6	6	7	7	7	7	7	7	5	5	5	4	4	4	2	54.8
TODAY TUE · THU · FRI		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2	49.1
IN THE TICKET OFFICE		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1	1		
ON THE FLOOR MON · WED		3	3	4	4	4	5	5	5	3	5	5	3	3	4	3	3	3	2	61.1
ON THE FLOOR TUE · THU · FRI		3	3	4	4	4	5	5	6	4	6	6	4	4	5	4	4	3	2	50.1
PROPOSED MON		5	7	7	7	7	7	7	9	8	7	7	7	7	7	7	7	5	3	25.8
PROPOSED TUE · THU		5	8	8	8	8	8	8	10	9	7	7	7	7	7	7	7	5	3	35.2
PROPOSED WED		4	7	7	7	7	7	7	9	9	7	7	7	7	7	7	7	5	3	30.0
PROPOSED FRI		4	7	7	7	7	7	7	9	9	8	8	8	8	8	8	8	6	3	26.4
IN THE TICKET OFFICE		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1	1		
ON THE FLOOR MON		4	6	5	5	5	5	5	7	4	5	5	5	5	6	6	6	4	3	32.5
ON THE FLOOR TUE · THU		4	7	6	6	6	6	6	8	5	5	5	5	5	6	6	6	4	3	41.9
ON THE FLOOR WED		3	6	5	5	5	5	5	7	5	5	5	5	5	6	6	6	4	3	35.4
ON THE FLOOR FRI		3	6	5	5	5	5	5	7	5	6	6	6	6	7	7	7	5	3	32.8
TRAIN CARRIAGES, DEC 2026	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66	

Saturday

staffed 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2	43.2
IN THE TICKET OFFICE		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1			
ON THE FLOOR		3	3	2	2	2	2	3	4	5	4	4	4	4	5	4	3	2	2	65.9
PROPOSED		4	6	6	6	6	6	6	7	9	7	8	8	8	8	8	8	5	4	24.0
IN THE TICKET OFFICE		1	1	2	2	2	2	2	2	4	2	2	2	2	1	1	1			
ON THE FLOOR		3	5	4	4	4	4	4	5	5	5	6	6	6	7	7	7	5	4	51.4
TRAIN CARRIAGES, DEC 2026	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41	

Sunday

staffed 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	FIT
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3	64.5
IN THE TICKET OFFICE			1	1	2	2	2	2	2	3	3	1	1	1	1	1	1	1		
ON THE FLOOR			2	3	2	2	2	2	3	5	5	4	3	3	3	3	2	2	3	71.7
PROPOSED			4	4	4	5	5	5	5	7	10	6	6	6	5	5	5	5	3	36.2
IN THE TICKET OFFICE			1	1	2	2	2	2	2	4	4	2	2	2	2	2	2	2		
ON THE FLOOR			3	3	2	3	3	3	3	3	6	4	4	4	3	3	3	3	3	34.9
TRAIN CARRIAGES, DEC 2026			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50	

Ticket office. Its staff are not floor cover, bar one of each pair at the quiet ends (until 08:00, or 09:00 Sunday; from 19:30; half on a Sunday evening). Today's office matches the plan except on **Sunday**: two earlies and **one** late (a 14:30–23:25 closer, in the office until 22:30), where the plan has two and two. The indented rows take the office out of each side, less its quiet-end help, so the floor rows compare like with like.

Cover weeks are not in these figures — a cover week has no fixed times — so real cover is a little higher. **Fit** is one number for how closely the people on duty follow the trains through the day: 0 would be a perfect match, lower is better. Page 1's weekday figure is worked out from the five weekdays combined, so it is not an average of the scores shown. On Sundays five trains in the December 2026 timetable arrive or leave after the 23:25 finish (the last at 23:54); the bar under the Sunday 23:00 hour marks the time after the finish. Whether Sunday cover should run later is still to be decided.

Reading it. The **floor** rows are the fair comparison. The ticket office is staffed to its opening hours, not to the trains, so counting it in can make a day look better or worse than the floor really is. The fit measures **shape, not numbers** — each hour's share of the day's staff against its share of the day's trains — so an extra person in a quiet hour makes it worse even though nobody is worse off. Read it beside the numbers of people, never alone.

Hard limits and the December 2026 rules

Hard limits first — met or not. Then the December 2026 rules, each met or not. The fatigue factors are on page 8.

Hard limits — a design either meets these or cannot be run

- ✓ **No more than 13 days worked in a row** — longest possible run **9** days (today 9)
Longest possible run is 9 days, within Chiltern's limit of 13. A cover week is 4 duties in 7 days and the roster clerk places them, so this is the worst case the link allows. Worked as a block of four rather than split day-on-day-off, the answer is the same — **9** days — wherever the block falls, so this row does not depend on how a cover week is placed.
The written source of the 13-day limit is still to be confirmed.
- ✓ **At least 12 hours between duties** — **0** rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today 0)
Tightest anywhere: **14h 05m — line 11 Thu 15:45-23:55 into Fri 14:00-22:30** (today's tightest: 12h 30m).
- ✓ **The contracted week, exactly** — **35h 00m** average Mon-Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sunday duties (83h 30m in all) are paid on top as rest-day working, as they are today. Individual weeks range 24h 00m–48h 30m; only the average is the contract.

The December 2026 rules — the same rules every proposal is measured against, with today's link beside them

RULE	TODAY'S LINK	THIS PROPOSAL	WHAT THE RULE ASKS
At least four on at the open, every day	× 4 · 4 · 3	✓ 4-5 · 4 · 4	from 06:20 (07:15 on a Sunday)
At least three through to the close, every day	× 2 · 2 · 3	✓ 3 · 4 · 3	until 23:55 (23:25 on a Sunday)
At least five still on duty at 22:00, every day	× 4 · 2 · 3	✓ 5-6 · 5 · 5	someone finishing at 22:00 does not count
Fourteen on a Saturday, ten on a Sunday	× 10 · 8	✓ 14 · 10	
Four cover weeks, evenly spread	× lines 1, 7, 12, 17	✓ lines 6, 12, 18, 24	evenly means every 6 lines, e.g. 1, 7, 13, 19
The ticket office: two early and two late, identical shifts, every day	× Mon-Fri yes · Sat yes · Sun no	✓ Mon-Fri yes · Sat yes · Sun yes	Mon-Fri 06:20-14:20 and 14:00-22:30; Sat 06:20-14:50 and 14:30-22:00; Sun two from 07:15 and two to 22:30
Every weekday closer starts at 15:45	× closers start at 15:15	✓ closers start at 15:45	
At least two on the floor at every moment	✓ fewest 2 · 2 · 2	✓ fewest 2 · 2 · 2	the ticket office counts only while its second person helps on the floor (until 08:00 or 09:00 on a Sunday, and from 19:30); checked every five minutes
Handovers: 15 minutes to each closer, 20 in the ticket office	✓ floor met · ticket office 20 min	✓ floor met · ticket office 20 min	on a Sunday each opener also stays until 15 minutes after the last closer arrives; where the design does not roster the office pairs, the plan's posts are assumed
Sunday duties between 8h and 9h	✓ 8h 00m-8h 55m	✓ 8h 00m-8h 30m	
No more shift times than today	✓ 18	✓ 15	today 18

Still to settle

- **Nothing to settle on the rules** — all 11 are met.
- **New shift times.** 10 of the 15 are times nobody works today; page 5 lists them.
- **Sunday's finish.** Five trains in the December 2026 timetable arrive or leave after the 23:25 Sunday finish; whether Sunday cover should run later is still to be decided.

ORR fatigue factors

From the ORR's good-practice guidance, *Fatigue Factors*, page 3 (December 2021). Δ present — the pattern is in this design and worth a look, not a breach · $\checkmark$ clear — it is not · $\bullet$ standing — comes with the station's hours or with any rotating link, not with this design's choices · – means it does not apply here.

These are 25 roster patterns that tend to tire people — 21 from the Office of Rail and Road's good-practice list and 4 checks from the rail industry's fatigue guidance (MRSF) — long runs of earlies, short gaps between duties, start times that jump about. For each one this table asks whether the pattern is in the rotation, today and proposed, and how big it is. The list is guidance: a factor present is a question to discuss, never a pass or a fail, and a design showing nothing is not thereby approved.

CODE	FACTOR	TODAY'S LINK	PROPOSED
FF1	Night shift covering 00:00–05:00 Time of day	–	–
FF2	Early shift starting 05:00–07:00 Time of day	$\bullet$ 24	$\bullet$ 27
FF3	Very early shift starting before 05:00 Time of day	$\checkmark$ 0	$\checkmark$ 0
FF4	Very early shift (before 05:00) over 8h Duty length	–	–
FF5	Day shift over 12h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF7	Early shift over 10h Duty length	$\checkmark$ 0	$\checkmark$ 0
FF8b	Fewer than 2 days rest after a block of early starts Recovery time	Δ 3	Δ 7
FF11	More than 13 consecutive shifts without a 48h break Recovery time	$\checkmark$ 12 $\checkmark$ 9 as rostered	$\checkmark$ 13 $\checkmark$ 11 as rostered
FF10	More than 4 consecutive 12h day shifts Cumulative	$\checkmark$ 0	$\checkmark$ 0
FF15	More than 4 consecutive early shifts in a rotating pattern Cumulative	Δ 6	$\checkmark$ 4
MRSF	More than 12 consecutive day shifts Cumulative	$\checkmark$ 9	$\checkmark$ 9
MRSF	More than 55 hours worked in any 7-day period Cumulative	Δ 60.8	Δ 58.0
MRSF	More than 7 consecutive 8h shifts Cumulative	Δ 9	$\checkmark$ 7
MRSF	More than 6 consecutive night or early shifts in a permanent pattern Cumulative	–	–
FF17	Backward rotating pattern Circadian	$\checkmark$ 10 backward / 12 forward	$\checkmark$ 8 backward / 18 forward
FF18	Rotating pattern of about a week Circadian	$\bullet$ 20-line rotation · typically 4h 00m a week	$\bullet$ 24-line rotation · typically 2h 26m a week
FF19	Successive start times varying by more than 2 hours (definition to confirm) Circadian	Δ 6.0	Δ 8.0
FF6	Night shift over 10h Night shifts	–	–
FF8	Less than 2 days rest after consecutive nights Night shifts	–	–
FF9	Less than 14h rest in any 24h for night shifts Night shifts	–	–
FF12	Only one day rest after night shifts Night shifts	–	–
FF14	More than 4 consecutive nights in a rotating pattern Night shifts	–	–
FF16	More than 3 consecutive night shifts over 8h Night shifts	–	–
FF20	First night shift Night shifts	–	–
FF13	Less than 12h rest in any 24h Intervals between duties	$\checkmark$ 0	$\checkmark$ 0

Design-specific findings: 3 (today 5). **Standing factors ($\bullet$) are recorded for information, not counted.** FF2 comes with the station's 06:20 open, and FF18 — the ORR's concern with a rotation that changes shift type about once a week (*Managing rail staff fatigue*, 7.68) — comes with every weekly link, which moves everyone one line a week by construction; its week-to-week step is shown as information. **How to read the table.** The figure beside each mark is the worst case found, for today's link as for the proposal, except where a row counts only the duties the rota fixes: the 55-hour row counts a cover week as no hours, and FF8b counts none of its days as early. "As rostered" under FF11 is the worst of the places a cover week's four duties could fall when worked together, which is usual. The MRSF rows are extra checks from the rail industry's guidance on managing staff fatigue, listed beside the ORR's own. FF2 and FF15 use the ORR's early — a start from 05:00 to 07:00, not the 11:00 used elsewhere on this sheet; the station opens at 06:20, so every design has FF2. FF11 counts shifts between two-day breaks, so a single rest day does not reset it — that is why it can be longer than the most days worked in a row. **Cover weeks.** A cover week's four duties can fall in different ways, and on this design that moves FF11 (13 if spread out, 10–11 if worked together). Where they fall is for the roster office. This page is an aid to discussion, not a fatigue risk assessment.

Where it stands among the 28

Every proposal on the same figures, with today's link at the foot. This sheet's row is highlighted.

Like with like. Every figure below was worked out the same way for every proposal, from its rota, by the same code — so any two rows can be compared directly.

PROPOSAL	RULES MET	FATIGUE FINDINGS	MOST DAYS IN A ROW	SHORTEST REST	FULL WEEKENDS OFF	FLOOR FIT			SHIFT TIMES	ALREADY WORKED TODAY
						MON-FRI	SAT	SUN		
All Clear	11 of 11	0	9	14h 05m	2	42.1	37.8	77.5	16	7
Familiar Nine	11 of 11	0	6	14h 20m	6	27.7	15.0	31.3	15	9
Right Away	11 of 11	0	6	14h 05m	6	26.5	14.6	28.9	17	5
Just Enough	11 of 11	3	9	14h 05m	2	34.3	51.4	34.9	15	5
Clean Sweep	10 of 11	3	9	13h 50m	2	34.6	40.0	53.5	18	11
Pinned Turns	8 of 11	0	6	14h 05m	6	35.8	21.5	60.8	16	9
Quarter To	8 of 11	0	6	13h 05m	6	49.7	43.7	93.9	13	12
Round Times	8 of 11	0	6	14h 05m	6	37.8	22.0	60.8	14	6
Weekend Capped	8 of 11	1	6	13h 35m	6	49.7	21.9	73.4	11	9
Same Turns	7 of 11	1	6	13h 30m	6	52.0	43.7	93.9	15	15
By the Book	5 of 11	0	6	13h 35m	6	35.0	11.1	39.4	19	0
Eight Forty	5 of 11	0	6	14h 20m	6	81.6	22.3	27.9	26	2
Gates Mended	5 of 11	1	6	14h 05m	6	38.4	42.0	49.0	15	4
Fifteen Turns	5 of 11	7	9	11h 15m	2	38.4	44.8	49.0	15	5
Office Written In	4 of 11	0	6	13h 50m	5	81.8	13.3	80.6	26	4
Light Retime	4 of 11	3	9	12h 30m	4	38.5	33.5	50.6	23	11
Saturday Four	4 of 11	3	9	13h 35m	5	38.5	35.8	64.8	18	10
Weekday Lates	4 of 11	5	9	13h 35m	6	40.4	20.7	64.8	18	13
Anchored Lines	3 of 11	1	7	14h 30m	6	41.1	20.7	64.8	19	13
Evening Peak	3 of 11	1	6	13h 05m	6	47.6	20.7	64.8	19	13
Frozen Block	3 of 11	1	6	14h 30m	4	41.1	20.7	64.8	19	13
Tenth Sunday	3 of 11	2	6	12h 30m	5	38.5	33.5	52.1	24	11
Cover at Seventeen	3 of 11	3	9	13h 35m	5	38.5	20.7	64.8	19	13
Short Closer	3 of 11	3	9	13h 35m	4	38.5	53.3	64.8	21	12
Targeted Fatigue Redo	3 of 11	4	9	13h 35m	5	38.5	20.7	64.8	19	13
Three Mondays	3 of 11	4	9	13h 35m	5	38.5	20.7	64.8	19	13
Weeks 17-18 Swapped	3 of 11	5	9	13h 35m	6	38.5	20.7	64.8	19	13
Polished Clean	2 of 11	4	9	13h 50m	2	38.7	51.4	72.4	19	9
<i>Today's 20-line link</i>	4 of 11	5	9	12h 30m	4	52.0	65.9	71.7	18	18

Grouped by rules met, then fewest fatigue findings; within a group, alphabetical — not ranked by any other figure, so read across the row. **Rules met** counts the 11 December 2026 rules on page 7. **Fatigue findings** are the design-specific patterns present at worst — FF2 and FF18, on every weekly link, are recorded on page 8 but not counted. **Most days in a row** is at worst, against Chiltern's limit of 13; **shortest rest** must be at least 12 hours, or the rota cannot be run. **Full weekends off** are out of 24 (20 for today's link). **Floor fit** is how closely the people on the floor follow the trains (the ticket office counted only while its second person helps at the quiet ends) — lower is closer (page 6). **Shift times** is how many different times the rota uses, and **already worked today** how many of them people work now.

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

Paste the whole block into the app's Links designer (Links → Import). It re-runs the grid, the hours, the hard limits, the fatigue factors and the hour-by-hour cover from the pasted days; the rest is worked out by the tools that made this sheet (see below).

The same rota is also supplied as two files, one to paste and one the app can open.

```

1, RD,          06:20-13:20, 06:20-13:20, 06:20-13:20, 06:20-13:20, 15:45-23:55, RD
2, 07:15-15:45, 06:20-13:50, RD,          RD,          06:20-13:50, 06:20-13:50, 06:20-14:50
3, RD,          RD,          06:20-13:50, 06:20-13:50, 06:20-13:50, 06:20-13:50, RD
4, 07:15-15:45, 06:20-14:20, 06:20-14:20, 06:20-14:20, 06:20-14:20, RD,          15:55-23:55
5, RD,          RD,          07:00-15:00, 07:00-15:00, 07:00-15:00, 07:00-15:00, 14:30-22:00
6, SP,          SP,          SP,          SP,          SP,          SP,          SP
7, RD,          RD,          07:00-15:00, 07:00-15:00, 07:00-15:00, 07:00-15:00, 06:20-14:50
8, 07:15-15:45, 07:00-15:00, RD,          RD,          07:00-15:00, 07:00-15:00, 07:00-15:00
9, 10:00-18:30, 13:00-21:30, 13:00-21:30, 13:00-21:30, RD,          RD,          06:20-14:20
10, RD,          13:00-21:30, 13:00-21:30, 13:00-21:30, 13:00-21:30, 13:00-21:30, RD
11, RD,          06:20-14:20, 06:20-14:20, 15:45-23:55, 15:45-23:55, 14:00-22:30, RD
12, SP,          SP,          SP,          SP,          SP,          SP,          SP
13, 14:00-22:30, 14:00-22:30, RD,          RD,          14:00-22:30, 14:00-22:30, 14:30-22:00
14, 15:25-23:25, 15:45-23:55, 15:45-23:55, RD,          RD,          15:00-23:00, 15:00-23:00
15, RD,          RD,          15:45-23:55, 15:45-23:55, 15:45-23:55, 15:45-23:55, 16:25-23:55
16, 15:25-23:25, 15:45-23:55, 15:45-23:55, 15:45-23:55, RD,          RD,          07:00-15:00
17, 14:00-22:30, RD,          RD,          RD,          06:20-14:20, 06:20-14:20, 15:55-23:55
18, SP,          SP,          SP,          SP,          SP,          SP,          SP
19, RD,          06:20-13:50, 06:20-13:50, 06:20-14:20, 13:00-21:30, 13:00-21:30, 13:00-21:30
20, RD,          RD,          14:00-22:30, 14:00-22:30, 15:45-23:55, 15:45-23:55, 15:55-23:55
21, 15:25-23:25, 15:45-23:55, RD,          RD,          RD,          06:20-14:20, 06:20-14:20
22, 07:15-15:45, 07:00-15:00, 07:00-15:00, 07:00-15:00, RD,          RD,          RD
23, RD,          14:00-22:30, 14:00-22:30, 14:00-22:30, 14:00-22:30, RD,          RD
24, SP,          SP,          SP,          SP,          SP,          SP,          SP

```

Prepared 28 September 2026. The code **49717d70** in every footer is the design's fingerprint — eight characters computed from these 168 days and nothing else by the tools that made this sheet (the app does not show it). If one day differs, the code changes. The grid, the hours, the hard limits, the fatigue factors and the hour-by-hour cover were calculated by the Marylebone Roster app (version 24.40) from exactly these days, and pasting them into the app shows the same figures there. The ticket office, the floor figures, the fits and the December 2026 rules are worked out from the same days by the proposal tools that made this sheet.